

ACME Inc. — Medallion Data Architecture

EDW Layer Design | Microsoft Fabric Lakehouse | June 2026

Architecture Overview

BRONZE — Raw	SILVER — Trusted	GOLD — BI-Ready
PURPOSE: Exact, immutable mirror of all source files. Full auditability. No transformations applied. SOURCES: 11 source files ingested as 12 Delta tables (CSV, XML, Excel). Shipments merged from 3 files. KEY FEATURE: _ingested_at timestamp on every table. Schema inferred at read time. TABLES (12): bronze_orders, bronze_order_details, bronze_shipments, bronze_customers, bronze_products, bronze_categories, bronze_divisions, bronze_shippers, bronze_suppliers, bronze_employees, bronze_offices, bronze_budget	PURPOSE: Single source of truth. Typed, cleaned, enriched, and validated. Business logic excluded. TRANSFORMATIONS: Dates parsed (M/d/yyyy). Geographic enrichment (Continent, SubRegion, ISO2). Known errors corrected. Unknown Members injected for referential integrity. GOVERNANCE: Invalid records quarantined in bronze_quarantine (not silently dropped). All fixes traceable. TABLES (12): silver_prefix for all source tables. Plus bronze_quarantine for rejected records.	PURPOSE: Star schema optimized for BI. Two fact tables share conformed dimensions (Constellation Schema). FACTS: fact_sales_items (17,032 rows, line grain), fact_orders (6,571 rows, order grain), fact_budget (40 rows, employee x year) DIMENSIONS: dim_date (2007-2021), dim_customer, dim_product, dim_employee, dim_shipper METRICS: SalesAmount, GrossMargin, GrossMarginPct, TotalOrderValue, BudgetMonthly, DeliveryDays AUDIT: gold_dq_issues - all DQ checks, BI-ready

Pipeline Workflow

1.	2.	3.	4.	5.	6.
Source Files	NB_01 Bronze	NB_02 Silver	NB_03 Gold	NB_04 DQ	Semantic Model
CSV/XML/Excel in OneLake	Delta tables. No transforms.	Typed, cleaned, enriched.	Star schema built.	9 checks. gold_dq_issues.	Direct Lake. Ready for BI.

Orchestration: Fabric Data Pipeline — notebooks execute in dependency chain. Each step runs only if previous completed successfully.

Implications of This Architecture

STRENGTHS

- Separation of concerns: raw data preserved independently of transformations
- Full auditability: any issue traced back to Bronze without re-ingestion
- Scalable: each layer evolves independently without breaking downstream
- Governance: quarantine framework makes DQ issues explicit and trackable
- BI-ready: Direct Lake on Delta tables eliminates import latency in Power BI

CONSIDERATIONS & NEXT STEPS

- Current pipeline is full-reload — production requires incremental (MERGE/UPSERT)
- GrossMargin uses current UnitCost — SCD Type 2 needed for historical accuracy
- Shipment SLA blocked by legacy date data (2007–2012 vs 2016–2021)
- DivisionID=2 PK violation requires source correction before go-live
- Unknown Members (Shippers 4 & 5, Employee -1) are interim — register in source systems